

Data Scientist at PIPET, The Catholic University of Korea. Hands-on in both [prompt engineering and evaluation frameworks for LLMs / generative AI](#) and [large-scale healthcare big data analysis and statistical modeling](#). Designed and ran the chatbot persona, LLM selection benchmark, and image-generation pipeline end-to-end for a K-pop virtual idol app. Covers the full pipeline from data extraction to analysis, modeling, and LLM applications with Python, SQL, and R.

SKILLS

|           |                                                                              |
|-----------|------------------------------------------------------------------------------|
| Languages | Python, R, SQL, SAS, SPSS                                                    |
| Libraries | scikit-learn, PyTorch, pandas, NumPy                                         |
| Tools     | Jupyter, RStudio, VS Code, Shiny, Tableau, Git, Linux, GitHub, Notion, Slack |

EXPERIENCE

**Data Scientist · Big Data Team, PIPET, The Catholic University of Korea** Feb 2024 – present

- Cohort extraction, preprocessing, and statistical modeling on nationwide HIRA healthcare big data (~50 million people) with SQL · R · SAS · Python
- Multiple myeloma (MM) — showed with causal inference and survival analysis that prior detection of precursor conditions improves survival; **published in Blood Cancer Journal (IF 11.6)**
- K-pop virtual idol app side project — designed and ran an **LLM benchmark, MBTI persona priming, and image-generation pipelines** end-to-end; documented in the [portfolio](#)

**Data Analyst · Medical Statistics Support Center, Inha University Hospital** Feb 2023 – Dec 2023

- Supported clinical research teams across departments with study design, data cleaning, statistical modeling, and manuscript statistics
- Completed **21 statistical analyses and 8 R training sessions**

EDUCATION

**M.S. in Statistics · Inha University** Mar 2021 – Feb 2023

Research: machine learning · neuroscience. Teaching assistant: Programming Languages, Data Structures, Introduction to Data Science, Statistical Analysis Methods & Data Mining, and more.

**B.S. in Statistics · Inha University** Mar 2016 – Feb 2021

SELECTED PROJECTS ★ [click a title for details](#)

LLM / AI Agent · Generative AI Pipelines (K-pop Virtual Idol App)

- LLM Model Benchmark** — shortlisted candidates from the EQ-Bench leaderboard → compared premium / lightweight top-5 models with cost simulation; evaluated response quality with MBTI priming and domain-curated questions. [\[chaeniverse/LLM-bench\]](#)
- MBTI Persona Priming** — reviewed an arXiv framework and designed a validation flow for LLM persona consistency: 16Personalities 60 items / 7-point Likert / 15 repetitions.
- Casting & Content Image Generation** — two stages (trainee casting, ~800 images; content scenes, ~3,200 images), each generated over a grid of prompt variables and reviewed qualitatively. Identity consistency via reference-image conditioning and negative constraints. [\[chaeniverse/kpop-image-experiments\]](#)
- OPT2I + DSG** — implemented the iterative prompt refinement algorithm of Mañas et al. 2024 (arXiv:2403.17804). DSG (Davidsonian Scene Graph) splits a prompt into small questions scored by VQA, quantifying how well an image reflects its prompt. [\[chaeniverse/opt2i-dsg\]](#)

Healthcare Big Data · ML

- Improved Survival in MM with Early Precursor Detection** — defined and extracted MGUS / SMM / de novo MM cohorts from ~50 million people in HIRA data; IPTW + marginal Cox regression showed the survival benefit of prior detection. **Blood Cancer Journal (IF 11.6, 2025)**. [\[chaeniverse/mm-precursor-survival\]](#)
- Long-term Opioid Use in MM** — defined and extracted 3,621 newly diagnosed MM patients from HIRA data; baseline risk-score model with low / intermediate / high-risk stratification. Manuscript in preparation (first author).
- OvaRisk-ML** — ovarian cancer risk classification with five ML models (RF, LR, XGBoost, CatBoost, MLP) + RFE + Bayesian HPO + Bayes-rule correction. **Deployed as a Python Shiny app** that outputs lifetime-risk cumulative incidence. Under review. [\[chaeniverse/OvaRisk-ML\]](#) [\[Shiny App\]](#)

Generative AI · Medical Imaging

- Future DaTScan Synthesis with cWDM** — conditional wavelet diffusion framework that synthesizes the 2-year follow-up image from a single baseline DaTScan; 746 longitudinal PPMI pairs. **Korean Statistical Society, first-author poster**. [\[chaeniverse/cwdm\]](#)
- Cross-Modality Neonatal Brain Image Conversion** — latent diffusion model synthesizing neonatal T1w from T2w MRI. **IEEE AIXMHC 2025 oral (co-corresponding author)**.
- Whole-Brain Functional Gradients in Migraine** — quantified brain function in migraine with non-linear manifold learning. **Human Brain Mapping (IF 4.7, 2023, first author)**. [\[chaeniverse/migraine\\_project\]](#)

Statistical Consulting · KPI Modeling

- Customer Satisfaction Index Modeling with SEM (Amos)** — modeled hospital service quality with structural equation modeling and computed a customer satisfaction index (CSI) from standardized regression weights. Consulting at the Medical Statistics Support Center, Inha University Hospital.